

Motivating Questions

- Williams identifies a tension between two strands of contemporary philosophy of science: (i) a “naturalistic” commitment to ground philosophical claims in our most successful scientific theories, and (ii) a widespread approach to theory interpretation centered on the counterfactual question: “*If this theory provided a true description of the world in all respects, what would the world be like?*”

Williams contrasts the counterfactual question with one closer to scientific practice: “*Given that this theory provides an approximately true description of our world, what is our world approximately like?*” (p. 210). How should we understand the difference between these two interpretive projects, and which should philosophers of physics pursue?

- Effective field theories (EFTs) become inapplicable beyond some short-distance scale and *incorporate this inapplicability into their mathematical framework*. Several philosophers of physics have argued that EFTs are therefore unfit for interpretation, or that they support at best an “opportunistic or instrumentalist attitude” (Butterfield & Bouatta). Williams argues the opposite: it is *only* by attending to EFTs that one can answer the “binding” interpretive question of Ruetsche (2011) – “does this interpretation allow the theory to discharge all of its scientific duties?”
- How does the EFT framework, together with the renormalization group (RG), enable a “divide and conquer” approach to scientific realism? What should a realist interpretation of an EFT be committed to, and on what grounds?

The Standard Account of Theory Interpretation

Williams characterizes “standard interpretation” by extracting five principles he takes to be at least tacitly shared by many in philosophy of physics (p. 212):

1. The theory is assumed to provide a *true and exhaustive* description of the physical world in all respects, including at all length scales.
2. The theory is to be interpreted *in isolation*: one may not appeal, e.g., to short-distance physics from a successor theory.
3. An interpretation identifies the set of all worlds nomologically possible according to the theory.
4. This set of worlds is determined by generic structural features (dynamical laws, kinematical constraints, symmetries); *information about how the theory is employed in practice is largely or entirely ignored*.
5. The goal is to identify the theory’s *fundamental* ontological features.

(Williams provides textual evidence to take this as something like conventional wisdom, although it might be hard to find any philosopher who endorses all of these claims as formulated.)

Corollary in the QFT case: the standard interpreter requires a *rigorous mathematical description at all length scales*. This restricts attention to toy models in 2 or 3 spacetime dimensions, since no mathematically rigorous version of any interacting QFT in $3 + 1$ dimensions has ever been constructed.

Vices of the Standard Account

- *Empirically applicable QFTs are declared unfit for interpretation.* In QFT, the question the standard interpreter actually sets out to answer is a nested counterfactual (p. 215): “If we lived in a world with two (or three) spacetime dimensions, and if that world could be described by a theory that we know is structurally incapable of describing our world, and if that theory provided a complete and exhaustive description of that world in all respects, what would that world be like?”
- *Loss of guidance from empirical applications.* How a theory is actually employed offers valuable information about which mathematical elements play a genuinely representational role and which are likely artefacts. Toy QFTs that lack empirical applications cannot provide this, and the standard interpreter cannot take advantage of this source of insight.
- *Reductio?* One can incorporate the short-distance breakdown of an EFT by placing the theory on a spatial lattice. Standard interpretation then treats the lattice as part of the theory’s description of the world. Since no one believes physical space has a lattice structure, Fraser concludes that EFTs are unfit for interpretation. Williams responds: any reductio leaves open which assumption to discard. An alternative diagnosis is that the approach to interpretation is unfit, not the theory.

Effective Realism: Divide and Conquer, Robustness

The positive proposal develops a “divide and conquer” (divide et impera) approach to realism, associated with Psillos and others. Two prongs:

- (i) Attend to the details of a theory’s empirical applications to distinguish *active* elements (those that play an essential role in the theory’s empirical success) from *idle constituents*.
- (ii) Identify theoretical elements that are stable and “robust” across independent ways of accessing them.

The criterion for ontological commitment is borrowed from Wimsatt: include in one’s ontology those entities, properties, or relationships that are *robust*, “*accessible (detectable, measurable, derivable, definable, producible, or the like) in a variety of independent ways*” (p. 95). Williams illustrates the criterion with Perrin’s argument for the existence of molecules. (Question: how closely does the EFT/RG case parallel the Perrin case? What plays the role of the multiple, independent experimental routes?)

The RG is supposed to play two crucial roles in carrying out the divide-and-conquer strategy in QFT: it both (a) explicitly delineates the length scales beyond which the theory becomes unreliable, and (b) identifies elements of the theory that are invariant across independent choices of how to model the short-distance physics. Both are versions of the robustness criterion. This is meant as a corrective to what Stachel (1993) called the “*fetishism of mathematics*”: “the tendency to assume that all the mathematical elements introduced in the formalization of a physical theory must necessarily correspond to something meaningful in the physical theory, and, even more, in the world that the physical theory purports to help us understand” (Williams p. 218).

Approximate Truth: Mirror Fermions and Emergent Lorentz Invariance

Two illustrations of robustness arguments in the EFT setting:

Mirror Fermions (Lattice Artefacts). The Nielsen–Ninomiya theorem implies that naïve “latticeizations” of a QFT containing fermions on a d -dimensional spacetime lattice yield 2^d times as many fermionic degrees of freedom as the target theory. (A single fermion field becomes sixteen on a four-dimensional lattice.) A

standard interpretation would seem committed to fifteen unwanted “mirror fermions” in the ontology. In more sophisticated latticization schemes, the action decomposes as

$$S_{\text{lattice}} = S_{\text{continuum}} + (a^p)S_{\text{mirror}},$$

where the mirror term is proportional to a positive power p of the lattice spacing a . The mirror fermions are sensitive to the arbitrary choice of latticization scheme and lattice spacing; $S_{\text{continuum}}$ is not. By the robustness criterion, the mirror fermions are *not* candidates for ontological commitment, even though they appear in the formalism.

Emergent Lorentz Invariance. Consider a scalar field on a four-dimensional spacetime lattice. The lattice violates Lorentz invariance at the “fundamental” (short-distance) level. The action has a Lorentz-invariant piece and a sum of symmetry-violating interactions multiplied by positive powers of a . RG analysis shows that at length scales $L \gg a$, the symmetry-violating interactions decay at a rate proportional to $(a/L)^p$, while the Lorentz-invariant structure remains. In the physical domain where the EFT is empirically reliable, the theory is fully Lorentz invariant – despite a “fundamental” description that strongly breaks the symmetries of special relativity.

Two morals Williams draws:

- Against Fraser (2009): one need not assign ontological significance to symmetry-violating short-distance modeling choices, since they are robustly absent from the empirically reliable regime.
- We can be *misled* by interpreting only the empirically unreliable “fundamental” structure of a QFT and conflating its content there with its content at all scales.

Scales and Ontology: QCD and Confinement

The second main argument of §3 (pp. 227–231): any interpretation of QCD that “enables the theory to discharge its scientific duties” must include higher-level entities in its ontology – not only the “fundamental” quark and gluon fields.

The QCD action $S_{\text{QCD}} = S_{\text{gluon}} + S_{\text{quark}} + S_{\text{interaction}}$ describes quark and gluon fields and their interactions. At short distances the strong coupling is small and particulate descriptions in terms of quarks and gluons are explanatorily powerful. But the strong coupling grows at longer distances, and beyond a length scale of roughly 10^{-15} m, the quark and gluon fields are *confined* into hadrons (protons, neutrons, pions, ...). At those length scales:

- It is *nomologically impossible* to separate a pair of particulate states of quark or gluon fields by more than the confinement scale.
- Hadrons have properties (no color charge, different symmetries, different dynamics) that quarks and gluons do not.
- Experimental particle physics (e.g., at the LHC) involves preparing, manipulating, and detecting hadrons. Explanations of scattering events make essential reference to hadronic degrees of freedom and their dynamics.

Williams’s claim: an interpretation of QCD that admits only quarks and gluons leaves the theory unable to discharge its explanatory duties. In support, he quotes Wimsatt (2007, p. 210):

The fact that most direct interactions of something at a level of organization will be with other things at that level means that detectors of entities at a level ... will interact with it via properties characteristic of that level ... For these reasons, and for others, eliminative reduction is often not possible, necessary, or desirable – our very instruments anchor us ... at the level we are observing.

Generalization: this story applies to any non-abelian gauge theory satisfying mild conditions – which includes the theories of the standard model. QCD “thus provides a case in which even though one knows that some higher-level entities are composite states of more fundamental entities, one cannot explain the behaviour of those higher-level entities solely in terms of the behaviour of the fundamental level” (p. 231).

Closing Themes

Williams’s positive view:

- A more discerning realism that abandons the standard interpreter’s “quixotic focus on fundamental structure” (p. 232).
- Ontological commitment guided by (i) detailed attention to how a theory is employed in practice, and (ii) robustness under variation in modeling choices, regularization schemes, hypothetical short-distance physics, and so on.
- Robust elements are precisely those that one expects to survive episodes of theory change. Given the inevitability of theory change in QFT (quantum gravity), this offers a more secure footing for realist commitment than the standard alternative.

The general moral (§4): “*the philosophical project of scientific realism will be better grounded, and more promising, if we introduce*” the resources Williams describes (p. 232).

Questions for Discussion

- How fairly does Williams characterize the “standard account”? Is this a straw man? Are the five principles a single coherent view, or a family of related but separable commitments? Which standard interpreters (if any) accept all five?
- Williams’s reframing of the interpretive question (counterfactual vs. “approximately true”) closely parallels themes from earlier in the seminar. How does Williams’s project connect to and depart from these?
- The robustness criterion (Wimsatt) plays an important role; is the criterion well enough specified to play this role? Compare with Perrin: there one has explicit, independent experimental routes converging on a single quantity. How can we characterize the situation for EFTs? Are RG-invariance under choice of regularization scheme and lattice spacing on a par with multiple independent experiments?
- What is the relation between Williams’s view and Wilson’s *Physics Avoidance*? Both criticize a “Theory *T*” style of interpretive practice that abstracts from applications; both stress scale-dependence and the failure of fundamentalism. Is Williams’s robustness-based realism compatible with Wilson’s broader picture, or are they in tension? (See also Wilson’s discussion of the “aspirational hope” vs. “physics avoidance” contrast.)